

# Reduced-Basis Steady Solver for EXHALE

*This memo develops a direct steady-state solution strategy for EXHALE that does not place the full, hours-long time-dependent calculation inside an MCMC loop. The proposed state is a smooth, low-dimensional expansion in radius. Mass conservation is imposed analytically, while momentum, energy, boundary, chemistry, radiation, and sonic-point conditions are enforced by scaled residuals. Bulk advection, viscous and conductive transport, elemental diffusion, and ion-stage advection are treated according to their conservative fluxes and current status in EXHALE. Two distinct formulations are documented: (1) a reduced basis evaluated by EXHALE's existing finite-volume residual, which targets the current discrete fixed point exactly, and (2) a conservative weak spectral-element formulation in which source terms are integrated by Gaussian quadrature. Nonlinear least squares is the primary solver. Annealed MCMC, sequential Monte Carlo, or posterior sampling is reserved for finding multiple branches and measuring degeneracies. This is a design document, not a claim that the proposed solver is already implemented or validated. The current production workflow and the reduced proposal are compared explicitly. Public numerical precedents are also surveyed, with AUTO-07p selected as the primary boundary-value reference and ESTER as the primary astrophysical reference.*

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## 1. Objective and scope

The computational problem is to obtain a physically correct stationary solution of the one-dimensional radiation-hydrodynamic equations solved by EXHALE without following the long transient required by explicit relaxation. The intended use is a single planet and a fixed set of physical inputs, not a Bayesian retrieval that reruns the complete hydrodynamic model at every sample.

Let the stationary equations be represented abstractly as

$$\mathcal{R}[\mathbf{U}; \boldsymbol{\theta}] = 0, \quad (1)$$

where  $\mathbf{U}(r)$  is the radial atmospheric state and  $\boldsymbol{\theta}$  contains fixed planetary, stellar, atomic, and boundary parameters. A conventional mesh-based Newton solve treats all values of  $\mathbf{U}$  as independent unknowns. The proposal instead restricts the state to a smooth manifold,

$$\mathbf{U}(r) = \mathcal{P}(r; \mathbf{a}), \quad \mathbf{a} \in \mathbb{R}^K, \quad K \ll 3N, \quad (2)$$

and searches for coefficients  $\mathbf{a}$  that make the governing residual small. The map  $\mathcal{P}$  must enforce positivity, composition normalization, and as much exact conservation as possible before any optimizer is called.

The method is related to weighted-residual, collocation, spectral, and spectral-element methods [8, 3, 17]. A residual interpreted as a statistical likelihood also resembles Bayesian physics-informed and probabilistic differential-equation methods [4, 19, 12]. The spline implementation proposed here is deliberately much smaller and more physically constrained than a general neural network.

The design has four goals:

1. reduce the nonlinear search from values at every radial cell to a few tens of coefficients;

2. preserve the conservative structure of the hydrodynamic equations;
3. retain EXHALE's existing radiation, ionization, heating, and cooling calculations rather than replace them with a fitted surrogate; and
4. distinguish numerical solution finding from posterior uncertainty quantification, so that an arbitrary likelihood width is not mistaken for physical uncertainty.

## 2. The current EXHALE fixed point

EXHALE inherits the Godunov finite-volume hydrodynamic core of ATES [2]. Its main marching path reconstructs left and right states, evaluates numerical fluxes and sources, and applies a three-stage strong-stability-preserving Runge–Kutta update. The relevant implementation is in

| file                                      | role                                                                    |
|-------------------------------------------|-------------------------------------------------------------------------|
| src/EXHALE_main.f90                       | three-stage hydrodynamic update and the optional steady-solver hand-off |
| src/modules/time_step/RK_rhs.f90          | spherical finite-volume flux divergence and hydrodynamic source terms   |
| src/modules/time_step/steady_residual.f90 | shared mass, momentum, and energy residual                              |
| src/modules/time_step/steady_newton.f90   | state packing, equilibrium radiation/chemistry refresh, PTC, and JFNK   |
| src/modules/wind_ae/wae_difeq.f90         | a separate steady BVP relaxation with base and sonic-point conditions   |

For a physical cell  $j$ , the present residual has the schematic form

$$R_{\rho,j} = (\nabla \cdot F_{\rho})_j - S_{\rho,j}, \quad (3)$$

$$R_{p,j} = (\nabla \cdot F_p)_j - S_{p,j} - S_{\mu,j}, \quad (4)$$

$$R_{E,j} = (\nabla \cdot F_E)_j - S_{E,j} - (\mathcal{H}_j - \mathcal{C}_j) - S_{E,\mu,j}, \quad (5)$$

where the optional  $\mu$  terms denote viscous work, molecular transport, and thermal conduction. The same reconstruction and flux calculation are used by time marching and by the steady residual. A true fixed point of the current discretization therefore satisfies the same equation that the explicit code relaxes.

Before evaluating Eq. (5), the full residual path converts the candidate conserved state to primitive variables, computes temperature, updates excited hydrogen when enabled, solves the ionization equilibrium, and recomputes heating and cooling. This detail is essential: reducing only the hydrodynamic unknowns does not freeze the microphysics. It eliminates the species fractions conditionally at every trial state.

The local Wind-AE implementation provides a second useful precedent. It solves a steady boundary-value problem directly, with density, temperature, velocity, ionization, and column densities as mesh unknowns. It imposes both

$\mathcal{M} = 1$  and the regularity numerator at the sonic point, then applies Newton relaxation. The reduced-basis proposal changes the representation and search algorithm, but the base-to-sonic BVP structure is already present in the source tree. Murray-Clay et al. [14] give the physical archetype for this class of transonic escape solution.

### 3. Comparison of the current and reduced solution strategies

The current EXHALE strategy and the proposed reduced strategy should not be viewed as mutually exclusive solvers. The production path is a conservative and robust way to approach a discrete attractor, while the reduced path is a candidate accelerator for reaching the neighborhood of a stationary root. A fair comparison must include the current hybrid workflow rather than describe EXHALE as time marching alone.

#### 3.1 What the current workflow actually solves

The production sequence is:

1. relax the finite-volume equations with three-stage SSP-RK updates;
2. when the mass-flux metric is sufficiently flat, optionally hand the full grid state to the JFNK steady solver;
3. evaluate equilibrium radiation, chemistry, heating, and cooling inside each nonlinear residual call;
4. include viscosity and thermal conduction through the same transport operator used by the marching path; and
5. when helium diffusion is active, alternate a steady hydrodynamic solve (JFNK from the marching hand-off with a budget of 500 iterations, or pseudo-transient continuation) with a relaxation of the element composition at that fixed wind, for at most 20 outer passes. The composition relaxation (`relax_element_composition`) steps on the composition time scale  $\min[\Delta r^2 / (D_{12} + K_{zz}), \Delta r / \max(|v|, w_s)]$  rather than the hydrodynamic CFL step, grows that step geometrically, and takes at most 400 steps. The outer loop is under-relaxed,  $X \leftarrow X_{\text{old}} + \omega(X_{\text{relaxed}} - X_{\text{old}})$ , starting at  $\omega = 0.5$  and halved to a floor of 0.125 on any pass whose drift failed to fall; it exits when the composition drift over a pass falls below  $10^{-3}$  of the base helium mass fraction.

The last operation is an outer co-convergence, not a monolithic hydrodynamics–diffusion Newton system. The JFNK residual contains mass, momentum, and energy but not the helium element-transport equation. Ion-stage advection is evaluated later by `post_process_adv.f90` and does not feed back into the converged equilibrium hydrodynamics.

The reduced proposal instead maps a small coefficient vector to the radial state, eliminates density by steady mass continuity where valid, refreshes the same microphysics, and solves a stationary residual directly. Depending on the formulation, that residual is either the existing EXHALE finite-volume residual, a strong collocation defect, or a conservative weak balance.

| property              | current EXHALE workflow                                                             | reduced-basis workflow                                                                                      |
|-----------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| unknowns              | conserved state in every radial cell, approximately $3N$ hydrodynamic unknowns      | typically tens of spline coefficients plus global quantities such as $\dot{M}$ and possibly a sonic radius  |
| route to stationarity | explicit relaxation followed by an optional full-grid PTC/JFNK finish               | direct nonlinear solve of the stationary coefficient residual, normally with continuation                   |
| conservation          | face-based finite-volume conservation using the production HLLC/WENO reconstruction | exact only for constraints built into the map or for correctly assembled conservative element residuals     |
| sharp structure       | robust numerical-flux treatment of shocks and narrow layers                         | smooth and efficient when resolved, but susceptible to missed layers or oscillatory basis error             |
| sonic branch          | approached through characteristic time evolution and polished by JFNK               | must be selected by explicit topology, regularity, and boundary constraints                                 |
| species diffusion     | implicit operator splitting and outer hydro-diffusion co-convergence                | can retain the production outer loop or add element fractions and flux equations to a simultaneous residual |
| ion-stage advection   | guarded post-processing without hydrodynamic feedback                               | requires additional transported fields or a conditional upstream solve                                      |
| discrete target       | unambiguously the production finite-volume fixed point                              | the same fixed point only in Formulation A; collocation and weak forms define different discrete problems   |

### 3.2 Advantages of the current EXHALE workflow

**Conservation and a defined fixed point.** The same reconstruction, numerical face flux, boundary routine, and source calculation are used in marching and in `steady_residual.f90`. A converged JFNK state therefore solves the equation that the production discretization relaxes. This supplies an unambiguous acceptance diagnostic.

**Robustness to localized and nonsmooth structure.** The finite-volume path is designed for steep fronts, contact modes, and shocks. It is safer than a coarse smooth basis when an ionization front, thermal layer, or diffusion scale height is narrower than the proposed basis support.

**A comparatively broad basin of attraction.** Time evolution follows causal propagation and physical relaxation. It can bring an imperfect initial atmosphere into the local convergence basin of JFNK. The code also returns to time marching if the JFNK finish fails, rather than accepting an unconverged Newton state.

**Reuse of established microphysics.** Radiation, chemistry, heating, cooling, viscosity, conduction, reconstruction, and boundaries remain on the normal production path. There is no second implementation whose agreement must be assumed.

### 3.3 Disadvantages of the current EXHALE workflow

**Long transients.** The calculation may spend most of its runtime evolving slow thermal, chemical, or diffusive modes even when only one stationary state is required. These hours-long trajectories are computational work toward the root, not additional information needed by a steady calculation.

**A high-dimensional nonlinear search.** The full-grid JFNK solve retains approximately  $3N$  hydrodynamic directions, including many radial perturbations that a smooth wind does not need. The resulting Jacobian can be poorly conditioned, and GMRES performance depends on the warm state and preconditioning.

**Local convergence of the Newton finish.** The production code deliberately waits until the mass-flux metric reaches the handoff neighborhood. JFNK accelerates the final convergence but does not currently replace the global relaxation stage.

**Split composition transport.** Helium diffusion is co-converged outside the JFNK residual, while ion-stage advection remains post-processing. This design is practical and internally documented, but it does not provide one Jacobian or one simultaneous residual for all transported composition fields.

**Limited branch tracing.** One time evolution normally selects one attractor. Parameter folds and several stationary branches are harder to map systematically than with pseudo-arclength continuation.

### 3.4 Advantages of the reduced strategy

**Removal of redundant search directions.** A smooth wind can often be represented by tens of coefficients rather than values in every cell. Exact mass-continuity elimination removes density as an independent function and gives  $\dot{M}$  a direct global meaning.

**Direct stationary solve.** The method seeks  $\mathbf{r}(\mathbf{a}) = 0$  without integrating the complete transient. It can therefore avoid the slowest relaxation modes. This is a potential speed advantage, not yet a measured EXHALE result: every trial may still require expensive nonlocal radiation and equilibrium chemistry.

**Constraints built into the representation.** Positive velocity and temperature, normalized composition, fixed base values, and exact steady mass continuity can be imposed before the nonlinear solver is called. This prevents exploration of many invalid full-grid perturbations.

**Natural continuation.** Radiation, chemistry, conduction, and diffusion can be activated progressively. AUTO-style pseudo-arclength continuation can also trace folds and distinguish subsonic, breeze, and transonic branches more systematically than a single relaxation trajectory.

**Potential simultaneous transport solution.** Adding a small abundance basis and a conservative total element-flux residual can place hydrodynamics and diffusion in one nonlinear system. This is useful only after the target energy and momentum closures have been stated; adding diffusive enthalpy or frictional heating would define physics beyond the current operator-split model.

### 3.5 Disadvantages and risks of the reduced strategy

**Basis truncation is a new model error.** A small scalar merit does not prove that a narrow heating, cooling, ionization, or diffusion layer is represented. Basis refinement, an independently located held-out residual, and comparison with the production-grid residual are all required.

**Conservation is not automatic.** A small strong residual at collocation nodes does not by itself guarantee an accurate integrated flux balance. Advective and diffusive contributions must enter one total conservative species flux. Weak conservation improves the integrated balance but no longer targets the exact HLLC/WENO fixed point.

**Sonic regularity becomes an explicit responsibility.** The reduced BVP can converge to a smooth subsonic breeze unless the intended branch and critical conditions are imposed. A differential equation divided by  $v^2 - c_s^2$  is singular at the sonic point and must be desingularized, split into domains, or accompanied by an explicit regularity equation.

**A small state does not imply a cheap residual.** Spline coefficients influence several cells, and column-dependent radiation couples the radial domain nonlocally. The reduced Jacobian is smaller but may be dense, while one residual evaluation may retain nearly the full chemistry cost.

**Smooth bases are unsuitable for genuine discontinuities.** A shock requires domain decomposition, an explicit shock location, or retention of the finite-volume solver. Regularization must not smooth away a physically required discontinuity merely to improve the scalar objective.

### 3.6 Recommended hybrid architecture

The safest use of dimension reduction is as a front end to the production solver:

$$\begin{aligned} &\text{simple Parker or frozen-source state} \longrightarrow \text{reduced continuation} \longrightarrow \text{reduced Formulation A solve} \\ &\longrightarrow \text{full-grid JFNK correction} \longrightarrow \text{short production evolution and residual check.} \end{aligned} \quad (6)$$

The first reduced prototype should use the existing finite-volume residual, not a newly derived continuum residual. It should solve only for  $v$ ,  $T$ , and  $\dot{M}$ , with density eliminated analytically and equilibrium chemistry refreshed conditionally. After that path is verified, adaptive collocation, weak conservation, and simultaneous elemental diffusion can be introduced in separate validation stages.

The current solver remains the final physical diagnostic and robust fallback. The reduced solver is successful if it places the state inside the production JFNK basin substantially faster while preserving the final EXHALE residual and physical branch. Wall-clock acceleration must be measured; it must not be inferred solely from the smaller coefficient count.

## 4. Two formulations that must not be conflated

There are two technically different ways to combine a reduced basis with EXHALE. They have different targets and different validation requirements.

### 4.1 Formulation A: reduced state, existing discrete residual

The coefficient vector is mapped onto all  $N$  EXHALE cells,

$$\mathbf{a} \xrightarrow{\mathcal{P}_N} \mathbf{U}_N(\mathbf{a}) \xrightarrow{\text{existing EXHALE evaluator}} \mathbf{F}_N(\mathbf{a}) \in \mathbb{R}^{3N}. \quad (7)$$

The reduced solve minimizes a scaled norm of  $\mathbf{F}_N$ . This formulation still evaluates every cell, but one evaluation is a stationary residual call, not a full time integration. It has three decisive advantages:

1. it targets the exact finite-volume fixed point of the current code;
2. all existing boundary, WENO3/HLLC, radiation, chemistry, and transport paths are reused; and
3. the full-cell residual exposes localized errors that a quadrature grid could miss.

It is the safest prototype because a failure cannot be attributed to a newly derived continuum residual. Gaussian quadrature is not needed in this first formulation.

### 4.2 Formulation B: conservative weak residual with quadrature

The second formulation defines a new spatial discretization. Divide the radial domain into  $N_e$  elements  $I_e = [r_e, r_{e+1}]$ . If  $A(r) = r^2$ , omitting the common solid-angle factor, a stationary conservation law reads

$$\frac{d}{dr} [A(r)\mathbf{F}(\mathbf{U})] = A(r)\mathbf{S}(\mathbf{U}, r). \quad (8)$$

Integrating over one element gives the conservative weak residual

$$\mathbf{R}_e = A_{e+1}\mathbf{F}(\mathbf{U}_{e+1}) - A_e\mathbf{F}(\mathbf{U}_e) - \int_{r_e}^{r_{e+1}} A(r)\mathbf{S}(\mathbf{U}, r) dr. \quad (9)$$

With  $Q$ -point Gauss–Legendre quadrature,

$$\int_{r_e}^{r_{e+1}} A\mathbf{S} dr \simeq \frac{\Delta r_e}{2} \sum_{q=1}^Q w_q A(r_{e,q}) \mathbf{S}[\mathcal{P}(r_{e,q}; \mathbf{a}), r_{e,q}]. \quad (10)$$

Equation (9) conserves the integrated flux exactly up to quadrature and nonlinear-solve errors. It does not require noisy numerical derivatives of the source terms. It is preferable to minimizing the squared pointwise differential residual when the atmosphere contains narrow heating, cooling, or ionization transitions. However, it no longer solves the exact HLLC/WENO3 finite-volume equation. Agreement with Formulation A should be expected only after basis, element, and quadrature convergence.

### 4.3 Recommendation

Implement Formulation A first. It tests whether dimension reduction itself removes the harmful search directions. Implement Formulation B only after the reduced manifold is demonstrated to contain the desired discrete solution. The two results should be reported as a discrete EXHALE fixed point and a weak continuum solution, respectively.

## 5. Low-dimensional representation

### 5.1 Radial coordinate

Map the physical domain to  $x \in [0, 1]$ . A logarithmic map is usually more appropriate than a linear map because most thermal and ionization structure is near the lower boundary:

$$x(r) = \frac{\ln(r/r_{\min})}{\ln(r_{\max}/r_{\min})}. \quad (11)$$

For an L1-truncated Roche domain, the map remains finite and monotone, but the outer boundary must be treated according to its actual characteristic structure; it must not be called a supersonic outflow boundary when the solution is subsonic there.

### 5.2 Velocity, temperature, and mass flux

For an outward solution with  $v > 0$ , use cubic B-splines

$$\ln v(x) = \sum_{k=1}^{K_v} a_k B_k(x), \quad (12)$$

$$\ln T(x) = \sum_{k=1}^{K_T} b_k B_k(x), \quad (13)$$

$$m = \ln \dot{M}. \quad (14)$$

The logarithms enforce positive velocity, temperature, and mass-loss rate. A single global Chebyshev expansion is attractive for very smooth Parker winds, but local B-splines are safer for EXHALE because their support is limited and knots can be concentrated near the base, a heating peak, an ionization front, or the sonic point. Global polynomials may oscillate across a narrow transition.

If a solution may contain inflow or a stagnation point,  $\ln v$  is invalid. One alternative is

$$v(x) = v_{\text{sc}} \sinh \left[ \sum_k a_k B_k(x) \right], \quad (15)$$

which permits a sign change without a hard kink. The mass-flux elimination below is then singular at  $v = 0$ , so a steady solution with nonzero  $\dot{M}$  cannot cross a true stagnation point. Such a state belongs to a different physical branch and should not be hidden by regularization.

### 5.3 Exact elimination of steady mass continuity

For a steady spherical single-fluid wind,

$$4\pi r^2 \rho(r) v(r) = \dot{M}. \quad (16)$$

Therefore

$$\rho(r; \mathbf{a}) = \frac{\exp(m)}{4\pi r^2 v(r; \mathbf{a})}. \quad (17)$$

Density is not assigned an independent spline. Equation (17) removes an entire function, prevents a flat but radially inconsistent mass flux, and gives  $\dot{M}$  a direct physical meaning. The factor  $4\pi$  can be omitted internally if EXHALE's code-unit flux is used, provided the convention is applied consistently.

This elimination assumes the mass equation has no distributed source. Any future mass loading, condensation, or separate-fluid transfer would invalidate Eq. (16); in that case the mass equation must return to the residual system.

### 5.4 Pressure and composition

Given  $\rho$ ,  $T$ , and equilibrium species fractions, pressure follows from the same equation of state used by EXHALE,

$$p = (n_{\text{tot}} + n_e) k_B T. \quad (18)$$

The coefficient-to-state map must call the existing composition and equation-of-state routines. A constant mean molecular weight would change the physical problem and would be acceptable only for a hydrodynamics-only prototype that is explicitly labeled as such.

The preferred design eliminates ion fractions through the existing equilibrium solver at every trial state. If an advected species must instead be represented, use logits. For stages  $s = 1, \dots, S$ ,

$$y_s(x) = \frac{\exp z_s(x)}{\sum_{t=1}^S \exp z_t(x)}, \quad z_s(x) = \sum_k c_{sk} B_k(x), \quad (19)$$

which enforces  $0 < y_s < 1$  and  $\sum_s y_s = 1$ . One logit should be fixed to zero to remove the redundant common offset.

### 5.5 Initial coefficient count

A practical first trial is

| quantity            | coefficients | treatment                                     |
|---------------------|--------------|-----------------------------------------------|
| $v(r)$              | 8–12         | cubic B-spline in $\ln v$                     |
| $T(r)$              | 8–12         | cubic B-spline in $\ln T$                     |
| $\dot{M}$           | 1            | scalar in $\ln \dot{M}$                       |
| $r_s$               | 0 or 1       | explicit only when a sonic constraint is used |
| $\rho(r)$           | 0            | eliminated by Eq. (17)                        |
| equilibrium species | 0            | solved conditionally at residual nodes        |

Thus the first nonlinear search has approximately 17–26 variables. Start with the coarsest basis that can represent a hot lower atmosphere and one sonic transition, then add knots by continuation. A small residual obtained only after adding many nearly redundant coefficients is evidence of poor scaling or an unresolved physical layer, not evidence that a larger MCMC is required.

## 6. Residual objective

### 6.1 Scaled physics merit

Let  $\mathbf{r}(\mathbf{a})$  collect hydrodynamic, boundary, sonic, and optional chemistry residuals. Define

$$\Phi(\mathbf{a}) = \frac{1}{2} \|\mathbf{W}\mathbf{r}(\mathbf{a})\|_2^2 + \Phi_{\text{prior}}(\mathbf{a}). \quad (20)$$

The diagonal or block matrix  $\mathbf{W}$  nondimensionalizes each residual. Without physical scaling, base momentum can dominate merely because gravity and pressure individually contain large cancelling terms, while an energy imbalance elsewhere is numerically smaller in code units.

For Formulation A, retain and extend EXHALE's cell-local state scales. A useful starting point is

$$s_{\rho,j} = \max(\rho_j, \rho_{\text{floor}})/t_{\text{dyn},j}, \quad (21)$$

$$s_{p,j} = \rho_j(|v_j| + c_{s,j})/t_{\text{dyn},j}, \quad (22)$$

$$s_{E,j} = \max(E_j, p_j, E_{\text{floor}})/t_{\text{dyn},j}, \quad (23)$$

with  $t_{\text{dyn},j}$  tied to the local crossing time. The exact scales must be tested by inspecting contributions from every equation and radius. No one row should become invisible because another row contains a large cancellation.

For Formulation B, scale an element residual by a characteristic flux through that element, not by its width alone:

$$\hat{R}_{k,e} = \frac{R_{k,e}}{\max(|A_e F_{k,e}|, |A_{e+1} F_{k,e+1}|, s_{k,e}^{\text{ref}})}. \quad (24)$$

Floors must represent a physical hydrostatic or thermal scale. A floor imported from an unrelated cell can distort the solution.

### 6.2 Boundary conditions

Boundary conditions may be imposed in either of two ways. A hard construction builds them into  $\mathcal{P}$ , for example

$$\ln T(x) = \ln T_0 + x \sum_k b_k \tilde{B}_k(x), \quad (25)$$

which guarantees  $T(0) = T_0$ . A soft condition adds

$$\Phi_{\text{BC}} = \frac{1}{2} \sum_i \left[ \frac{B_i(\mathbf{U})}{s_{B_i}} \right]^2. \quad (26)$$

Hard conditions reduce dimension and avoid weight tuning, but they should be used only for genuinely fixed physical conditions. A hard base velocity would conflict with EXHALE's valve or mass-flux condition. The base density and temperature, pressure-gradient consistency, and outer characteristic condition must be copied from the selected EXHALE configuration rather than inferred from an obsolete memo.

### 6.3 Sonic-point conditions

The conservative equations are not algebraically singular at  $\mathcal{M} = 1$ , but the transonic branch is selected by regularity. If the sonic radius  $r_s$  is an unknown, add

$$R_{\mathcal{M}} = \mathcal{M}(r_s) - 1, \quad (27)$$

$$R_{\text{crit}} = N_{\text{wind}}(r_s), \quad (28)$$

where  $N_{\text{wind}}$  is the numerator of the differential wind equation after the sonic denominator has been separated. The sonic merit is

$$\Phi_s = \frac{1}{2} \left( \frac{R_{\mathcal{M}}}{s_{\mathcal{M}}} \right)^2 + \frac{1}{2} \left( \frac{R_{\text{crit}}}{s_{\text{crit}}} \right)^2. \quad (29)$$

This mirrors the Mach-one and numerator conditions in `wae_difeq.f90`. For a domain whose physical solution is subsonic at L1, forcing an interior sonic point is wrong. Branch topology must be chosen before the constraint is enabled.

### 6.4 Smoothness prior

Spline truncation already imposes smoothness. If additional regularization is needed, use curvature rather than coefficient magnitude:

$$\Phi_{\text{smooth}} = \frac{\lambda_v}{2} \int_0^1 \left[ \frac{d^2 \ln v}{dx^2} \right]^2 dx + \frac{\lambda_T}{2} \int_0^1 \left[ \frac{d^2 \ln T}{dx^2} \right]^2 dx. \quad (30)$$

The same Gaussian quadrature can evaluate these integrals. Regularization may stabilize an underdetermined coarse solve, but it must not erase a physically required ionization or cooling front. A residual decrease obtained by increasing  $\lambda$  is not convergence of the governing equations.

### 6.5 Held-out residual

Never assess the weak solution only at the quadrature nodes used in Eq. (10). Evaluate a held-out residual on a denser, independently located grid. A low training residual and a large held-out residual indicate aliasing or a basis that oscillates between nodes. For Formulation B, also double  $Q$  without refitting and require the integrated balances to remain stable.

## 7. Radiation and chemistry

### 7.1 Conditional elimination

At a trial  $v(r)$ ,  $T(r)$ , and  $\dot{M}$ , the density is known from Eq. (17). The preferred residual evaluation is then:

1. construct  $\rho$ ,  $v$ , and  $T$  at all required nodes;
2. compute cumulative columns and attenuated radiation consistently;
3. solve the coupled equilibrium species system at those nodes;

4. recompute  $n_{\text{tot}} + n_e$ , pressure, excited-state populations, heating, and cooling; and
5. assemble hydrodynamic residuals from this self-consistent state.

This is a nonlinear variable-elimination or variable-projection strategy. It keeps chemical unknowns out of  $\mathbf{a}$  while retaining their response to the candidate hydrodynamic state.

## 7.2 Nonlocal columns

Photoionization couples radii through outward columns. Formulation A should use the current EXHALE column calculation unchanged. Formulation B has two choices:

1. compute each column by cumulative Gaussian quadrature from the outer boundary; or
2. introduce column fields satisfying

$$\frac{dN_s}{dr} = -n_s(r), \quad N_s(r_{\text{max}}) = N_{s,\text{max}}, \quad (31)$$

and add their weak residuals.

The first choice has fewer unknowns and is the recommended start. The second choice localizes the radiation equations and may provide a sparser Jacobian, but it adds at least one field for every opacity carrier.

## 7.3 Determinism requirement

The present full residual updates the species fraction array in place. This is efficient inside Newton continuation, but a statistical likelihood must be a deterministic function of  $\mathbf{a}$ . If the same  $\mathbf{a}$  yields different residuals depending on the previous proposal's chemistry, detailed balance is violated and even deterministic optimization can acquire a false noise floor.

Each trial must therefore use one of the following policies:

1. restore the same reference species state before every evaluation and converge equilibrium tightly;
2. prove by repeated evaluations that the local equilibrium is unique and independent of its initial guess; or
3. include the relevant chemical branch in the state vector explicitly.

Cache keys must include every state variable on which a rate depends. Caching based only on radius or column would make coefficient perturbations return stale physics.

## 7.4 Differentiability

Hard switches, clipped fractions,  $\max(v, 0)$  valves, table boundaries, and fallback chemistry solvers make  $\mathbf{r}(\mathbf{a})$  nonsmooth. This matters more for Gauss–Newton or Hamiltonian Monte Carlo than for time marching. The reduced solver should use the existing smooth valve where physically permitted, record every branch switch, and reject proposals that trigger an unconverged chemistry result. Smoothing a switch is acceptable only when its validity range and effect on the physical solution are documented.

## 8. Advection and diffusion in the reduced formulation

The reduced method remains applicable when advection and diffusion are active. They do not require a return to one unknown at every cell, but they change which fields, boundary conditions, and conservation equations must be retained. The essential rule is to reduce the state representation, not the conservative transport law.

### 8.1 Transport already present and transport still outside the fixed point

The present EXHALE paths must be distinguished before defining the target residual:

| process                          | present treatment                                                                                                       | implication for a reduced solve                                                                                              |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| bulk hydrodynamic advection      | conservative finite-volume mass, momentum, and energy fluxes                                                            | already included in Formulation A; retain conservative fluxes in Formulation B                                               |
| viscosity and thermal conduction | common operator used by marching and the steady residual when enabled                                                   | already included in the current discrete fixed point                                                                         |
| elemental diffusion              | implicit operator-split update, followed by outer hydrodynamics–diffusion co-convergence in the current Newton workflow | not a row of the JFNK residual; a direct simultaneous solve needs an additional conservative element-transport residual      |
| ion-stage advection              | guarded post-processing calculation                                                                                     | does not feed back into the current equilibrium hydrodynamic solution; simultaneous coupling defines an enlarged fixed point |

Thus, Formulation A already includes bulk advection, viscosity, and conduction through the production residual. It reproduces the fully co-converged result with elemental diffusion only if the same outer iteration is retained. It does not become a simultaneous diffusion or ion-advection solve merely by evaluating `steady_residual.f90`.

### 8.2 Bulk advection, viscosity, and conduction

Bulk advection is not a new free function. It is contained in the conservative hydrodynamic flux  $\mathbf{F}(\mathbf{U})$ , while steady mass continuity still permits the exact elimination in Eq. (17). In a continuum weak formulation, viscosity and thermal conduction enter through the total momentum and energy fluxes. Schematically,

$$F_p^{\text{tot}} = F_p^{\text{adv}} + p - F_p^{\text{visc}}, \quad (32)$$

$$F_E^{\text{tot}} = F_E^{\text{adv}} + F_E^{\text{cond}} + F_E^{\text{visc}}, \quad (33)$$

with signs fixed by the production convention. The weak residual uses these total fluxes in the boundary term of Eq. (9); the flux derivatives should not be reintroduced as unrelated pointwise sources. Cubic B-splines provide the first derivatives needed by standard constitutive laws and are normally sufficient. Sharp conductive fronts still require local knot or element refinement.

### 8.3 Elemental diffusion as a conservative flux constraint

When helium or a metal element separates from hydrogen, a constant elemental abundance is no longer a valid conditional variable. Introduce a bounded, low-dimensional elemental fraction, for example

$$\text{logit}\left(\frac{f_{\text{He}}}{f_{\text{max}}}\right) = \sum_{k=1}^{K_{\text{He}}} c_k B_k(x). \quad (34)$$

Six to ten coefficients are a reasonable initial experiment, but the accepted number must be determined by residual and basis refinement rather than fixed in advance.

The helium transport implemented by EXHALE can be represented schematically by one total flux,

$$J_{\text{He}} = n_{\text{He}}v - n_{\text{H}}D_{\text{He}} \left[ \frac{df_{\text{He}}}{dr} + f_{\text{He}} \left( G_{\text{He}} + \alpha_T \frac{d \ln T}{dr} \right) \right] - n_{\text{H}}K_{zz} \frac{df_{\text{He}}}{dr}, \quad (35)$$

where  $G_{\text{He}}$  collects the gravitational and ambipolar separation terms,  $D_{\text{He}}$  is the molecular diffusion coefficient, and  $K_{zz}$  is the eddy diffusion coefficient. The exact code expression, including its effective mass and sign conventions, must be called or factored from `src/modules/functions/binary_element_diffusion.f90`; Eq. (35) is not a replacement implementation. Note that the code does not transport the ratio  $f_{\text{He}} = n_{\text{He}}/n_{\text{H}}$  that Eq. (35) is written in: the transported variable is the helium MASS FRACTION  $X = \rho_{\text{He}}/\rho$  of a binary H/He mixture, and the diffusive mass flux is  $J = -\rho(D_{12} + K_{zz})dX/dr - \rho D_{12}GX(1-X)$  with  $G = [(m_{\text{He}} - m_{c1})m_{\text{H}}g - (\bar{Z}_{\text{He}} - \bar{Z}_{c1})eE]/(kT) + \alpha_T d \ln T/dr - d \ln \psi/dr$ .  $X$  is bounded at both ends of the composition axis, unlike  $f_{\text{He}}$ , which diverges in exactly the helium-rich limit the module exists for;  $m_{c1}$ ,  $\bar{Z}_{c1}$  and  $\psi$  are carrier quantities, so the same expression holds in the atomic, triplet-helium and molecular regions. Any reduced formulation must reproduce that flux, not Eq. (35). Multispecies escape and helium fractionation models provide the broader physical context for coupled advective–diffusive fluxes [11, 18, 16].

If chemical reactions conserve helium nuclei, steady elemental conservation is

$$\frac{1}{r^2} \frac{d}{dr} (r^2 J_{\text{He}}) = 0. \quad (36)$$

For an element  $e = [r_L, r_R]$  and a test function  $w_j$ , its weak residual is

$$R_{\text{He},e,j} = [w_j r^2 J_{\text{He}}]_{r_L}^{r_R} - \int_{r_L}^{r_R} \frac{dw_j}{dr} r^2 J_{\text{He}} dr. \quad (37)$$

Gaussian quadrature evaluates the remaining integral. With a constant test function, Eq. (37) is simply the difference between the total element fluxes at the two faces. This is the most transparent conservation diagnostic.

Diffusion also requires boundary conditions that pure hydrodynamics does not. The present physical choice is a fixed elemental ratio at the base and zero *diffusive* flux at the outer boundary. The latter must not set the total flux to zero: the outward advective flux remains allowed. Applying separate residuals to advective and diffusive divergences can create a numerical source or sink; they must be combined into  $J_{\text{He}}$  before conservation is tested.

### 8.4 Ion-stage advection

Local ionization equilibrium can be eliminated conditionally only when reaction times are short compared with transport times. If ion stages are advected self-consistently, each retained stage satisfies

$$\frac{1}{r^2} \frac{d}{dr} (r^2 J_{X^i}) = P_{X^i} - L_{X^i}, \quad (38)$$

where  $J_{X^i}$  includes the selected advective and diffusive contributions. Its weak residual is

$$R_{X^i,e,j} = \left[ w_j r^2 J_{X^i} \right]_{r_L}^{r_R} - \int_{r_L}^{r_R} \frac{dw_j}{dr} r^2 J_{X^i} dr - \int_{r_L}^{r_R} w_j r^2 (P_{X^i} - L_{X^i}) dr. \quad (39)$$

The production and loss integrals are evaluated at Gaussian nodes using the same attenuated radiation and thermochemical state as the hydrodynamic source terms.

Two reductions are practical:

1. Represent  $S - 1$  independent stage logits with splines and obtain all  $S$  fractions through a softmax map. This supports a coupled boundary-value solve and preserves positivity and normalization, but adds coefficients.
2. For a trial hydrodynamic and elemental state, integrate the ion-advection equations conditionally from the upstream boundary. This uses fewer global coefficients and is attractive for a strictly outward flow with  $v > 0$ .

The second option is not generally valid through stagnation or inflow, because the upstream boundary changes. Such cases require a boundary-value treatment or a conservative flux residual with characteristic boundary conditions.

Do not impose all ion-stage equations together with an independent equation for their sum. For one element, use one elemental transport equation and only  $S - 1$  independent stage equations. When reaction bookkeeping exactly conserves nuclei, the sum of the stage source terms is zero and the summed stage residual must reproduce the elemental residual. This identity is both a rank test and a stringent implementation diagnostic.

## 8.5 Energy consistency and the definition of the target problem

A fully coupled multicomponent model may require more than composition fluxes. Species diffusion can carry enthalpy,

$$F_E^{\text{species}} = \sum_s h_s J_s^{\text{diff}}, \quad (40)$$

and interspecies friction can heat the gas. Omitting these terms while adding composition diffusion is an approximation whose validity range must be stated. If the goal is the current EXHALE result, retain the present operator-split physics and reproduce its outer co-convergence. Adding Eq. (40) or new friction terms defines a different physical model and requires independent conservation tests.

For a simultaneous reduced transport solve, the residual therefore has the block structure

$$\mathbf{r}_{\text{total}} = \begin{bmatrix} \mathbf{r}_{\text{mass}} & \mathbf{r}_{\text{momentum}} & \mathbf{r}_{\text{energy}} & \mathbf{r}_{\text{element}} & \mathbf{r}_{\text{ion}} \end{bmatrix}^T. \quad (41)$$

Blocks are included only for fields that are solved dynamically. Equilibrium ion fractions must not also receive advection residuals. The extra spline coefficients remain compatible with optimization and MCMC because their number scales with the number of transported fields, not with the number of radial cells.

## 9. Optimization and Bayesian interpretation

### 9.1 Why MCMC is not the primary root finder

If the equations and boundary conditions define one deterministic solution, the target is  $\mathbf{r}(\mathbf{a}) = 0$ . Nonlinear least squares uses residual magnitudes and derivatives to approach that root. A random-walk MCMC sampler continues to explore after it has found the basin and therefore spends most evaluations estimating a distribution that was introduced artificially.

The recommended hierarchy is:

1. obtain a coarse solution by trust-region least squares or Levenberg–Marquardt;
2. continue in basis size, radiation coupling, or source strength;
3. use multistart, differential evolution, or annealed sampling only when several basins are plausible; and
4. sample a posterior only after a stable minimum and a defensible error model have been established.

### 9.2 Pseudo-posterior for branch exploration

A residual-based distribution can be defined as

$$\pi_\beta(\mathbf{a}) \propto \exp[-\beta\Phi(\mathbf{a})] p_0(\mathbf{a}), \quad (42)$$

where  $p_0$  supplies bounds and smoothness information. Increasing  $\beta$  concentrates the distribution around minima. Parallel tempering or sequential Monte Carlo can move between branches more effectively than one cold chain. The final modes provide starting points for deterministic root solves.

This construction is a search distribution, not automatically a physical posterior. Its width depends on  $\beta$ , residual scaling, element count, and regularization. Bayesian physics-informed models make a similar residual-likelihood construction [19], while probabilistic ODE/BVP methods place probability on solution functions and discretization error [4, 12]. Those frameworks justify uncertainty only after the stochastic model has been specified; the exponential form alone does not do so.

### 9.3 A physical posterior

A defensible posterior requires uncertain physical inputs, model discrepancy, or observational data. One possible hierarchy is

$$p(\mathbf{a}, \boldsymbol{\theta} \mid D) \propto p(D \mid \mathcal{G}[\mathcal{P}(\mathbf{a}), \boldsymbol{\theta}]) p(\mathbf{r}(\mathbf{a}, \boldsymbol{\theta}) \mid \Sigma_{\text{model}}) p_0(\mathbf{a}, \boldsymbol{\theta}), \quad (43)$$

where  $D$  is an observed spectrum,  $\mathcal{G}$  is the transit model, and  $\Sigma_{\text{model}}$  has a stated interpretation. Hydrodynamic escape analyses have used MCMC with grids or fast Parker-wind models to infer temperature, composition, and mass loss [13, 7]. Those are inference examples, not examples of replacing a hydrodynamic root solver with MCMC.

## 9.4 Sampler choice

For  $K \lesssim 30$ , an affine-invariant ensemble sampler is easy to deploy and parallelize [9], but it does not solve multimodality by itself. Hamiltonian Monte Carlo can be much more efficient in a smooth correlated basin, but requires reliable gradients. Function-space MCMC methods become relevant only if basis refinement is taken toward a high-dimensional random function [5]. For the proposed small spline expansion, careful scaling and a deterministic optimizer are more important than a specialized infinite-dimensional sampler.

Convergence reporting must include more than acceptance fraction. At minimum report independent-chain agreement, integrated autocorrelation times, effective sample sizes, trace plots for  $\ln \hat{M}$  and residual components, mode occupancy, and repeated likelihood determinism tests.

## 10. Jacobians and computational cost

Let  $\mathbf{J}_a = \partial \mathbf{r} / \partial \mathbf{a}$  be the reduced Jacobian. Even though  $\mathbf{r}$  may have thousands of rows,  $\mathbf{J}_a$  has only  $K$  columns. A forward finite-difference Jacobian costs  $K + 1$  residual evaluations and is a reasonable prototype for  $K \sim 20$ . Central differences double that cost. Colored differencing of mesh variables does not directly help because each spline coefficient has support over several cells, although compact B-splines preserve partial sparsity.

Better long-term options are:

1. analytic derivatives of the basis map combined with directional derivatives of the existing residual;
2. complex-step differentiation for code paths that contain no comparisons or nonanalytic functions of the perturbed quantity;
3. algorithmic differentiation after the equilibrium and table interfaces have been made compatible; or
4. matrix-free products  $\mathbf{J}_a \delta \mathbf{a}$  in a reduced Gauss–Newton or trust-region method.

If one full equilibrium residual evaluation costs  $C_R$ , a finite-difference Gauss–Newton iteration costs approximately  $(K + 1)C_R$ , plus a small dense least-squares solve. This must be measured directly. The hours-long time integration does not determine  $C_R$ ; chemistry and nonlocal radiation are expected to dominate a single call, but no runtime claim should be made before profiling.

Warm-starting chemistry can reduce  $C_R$  during deterministic continuation. For MCMC it is permitted only if it changes iteration count but not the final residual. A history-dependent approximate chemistry result is not an acceptable speed optimization.

## 11. Proposed algorithms

### 11.1 Algorithm A: exact discrete reduced solve

1. Select the physical branch and boundary configuration. Decide whether the domain contains a transonic wind or remains subsonic at its outer edge.

2. Choose coarse spline knots and initialize  $\ln v$ ,  $\ln T$ , and  $\ln \dot{M}$  from a cold, transonic, Wind-AE, or previously converged profile.
3. Evaluate the splines on the existing EXHALE cell centers.
4. Compute  $\rho$  from Eq. (17), then obtain pressure from the equilibrium composition and equation of state.
5. Pack the conserved state and apply the same ghost-cell boundary routine used by the production code.
6. Restore a deterministic chemistry initial state and evaluate the full EXHALE steady residual.
7. If elemental diffusion is part of the target, either complete the current outer hydrodynamics–diffusion co-convergence or append the conservative element residual in Eq. (37). If ion-stage advection is part of the target, append only the independent stage equations in Eq. (39).
8. Scale every residual row and minimize Eq. (20) by a trust-region least-squares method.
9. If progress stalls, inspect the largest physical residuals before adding knots. Do not infer that the basis is too small from the scalar merit alone.
10. Increase the basis dimension by knot insertion and prolong the old spline into the new basis. Continue until the solution and held-out residual are stable.
11. Pass the final state through the unmodified EXHALE residual diagnostic and, as a separate test, a short time-dependent relaxation.

## 11.2 Algorithm B: weak spectral-element solve

1. Start from the converged or best available result of Algorithm A.
2. Define elements independently of the spline knots. Place additional element boundaries near rapid thermal or ionization changes.
3. At Gaussian nodes, evaluate the spline state, conditional or transported chemistry, radiation, heating, cooling, and transport coefficients.
4. Assemble Eq. (9) for momentum and energy and, when active, Eqs. (37) and (39) for composition. Mass is already exact by construction.
5. Add boundary and branch-appropriate sonic constraints.
6. Minimize the scaled weak merit, then repeat with larger  $Q$ , more elements, and more spline knots.
7. Evaluate the continuous differential residual on an independent dense grid and the EXHALE discrete residual after projection onto its mesh.
8. Attribute differences correctly: quadrature error, basis truncation, continuum-versus-HLLC discretization, or a different boundary condition.

### 11.3 Algorithm C: mode search and posterior sampling

1. Construct the deterministic, scaled objective and verify repeated evaluations first.
2. Run many inexpensive deterministic starts or an annealed population over the coarsest basis.
3. Cluster minima by  $\dot{M}$ , sonic radius, temperature structure, and equation-wise residual, not by coefficient distance alone.
4. Refine every distinct physical mode with Algorithm A or B.
5. Only then define  $\beta$  or a physical covariance and run MCMC.
6. Confirm each sampled mode by an independent deterministic solve.

## 12. Relevant public codes and transferable methods

No public code identified in this survey combines all features proposed here: a reduced representation of a stationary astrophysical wind, the existing EXHALE radiation and chemistry, conservative species transport, and direct solution of the resulting coefficient residual. There are, however, mature public precedents for the important numerical components. The closest reference for the one-dimensional boundary-value algorithm is AUTO-07p. ESTER is the closest astrophysical precedent for a nonlinear spectral solve with complex microphysics. Dedalus and KADATH provide useful independent examples of spectral equation assembly, but they are lower-priority implementation references for EXHALE.

| code     | relevant capability                                                                                                                                       | principal mismatch with EXHALE                                                             |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| AUTO-07p | adaptive piecewise-polynomial collocation for ODE boundary-value problems; boundary and integral constraints; continuation and bifurcation analysis       | local ODE interface; no radiation–chemistry model or existing finite-volume fixed point    |
| ESTER    | nonlinear Newton solution of rotating-star structure with radial Chebyshev expansions, spherical harmonics, coordinate mappings, and stellar microphysics | smooth two-dimensional stellar structure rather than a transonic irradiated wind           |
| Dedalus  | open spectral framework for initial-value, boundary-value, and eigenvalue problems; symbolic equation and boundary-condition assembly                     | adopting it as the main solver would require a separate implementation of EXHALE physics   |
| KADATH   | multidomain spectral representation and nonlinear equation systems in general relativity and theoretical physics                                          | primarily elliptic equilibrium problems rather than conservative advection–diffusion winds |

## 12.1 AUTO-07p: the primary numerical reference

AUTO-07p solves ODE boundary-value problems by orthogonal collocation with piecewise polynomials. Its public manual specifies two to seven collocation points in each mesh interval and adapts the mesh to equidistribute a local discretization-error estimate. It also treats boundary conditions, integral constraints, parameter continuation, folds, and branch switching [6]. These features align directly with the reduced EXHALE problem:

$$\mathbf{y}(r) = (\ln v, \ln T, \text{logit } f_{\text{He}}, N_{\text{HI}}, N_{\text{HeI}}, \dots), \quad (44)$$

while  $\ln \dot{M}$ , a sonic radius, irradiation strength, or a diffusion multiplier can be treated as global continuation parameters. Nonlocal columns become local auxiliary equations through Eq. (31).

The most transferable AUTO-07p ideas are:

1. a local polynomial representation rather than one high-degree global polynomial;
2. an adaptive interval mesh driven by a defect estimate;
3. simultaneous enforcement of interior equations, physical boundary conditions, and global integral constraints;
4. pseudo-arclength continuation through parameter folds; and
5. explicit branch labels and restart states rather than interpreting every converged root as the same physical solution.

Continuation is especially relevant to EXHALE. A solution can first be found with frozen heating and composition, then followed as radiation, chemistry, conduction, or diffusion is turned on. This is generally safer than attempting the fully coupled root from an arbitrary initial profile.

AUTO-07p cannot accept a singular sonic equation without reformulation. The sonic point must be handled by a regularity condition, a desingularized independent coordinate, or domain splitting. Likewise, calling the complete EXHALE finite-volume and nonlocal microphysics path from an AUTO callback would obscure which discrete problem is being solved. The recommended use is to adopt its collocation, adaptivity, and continuation design in the EXHALE wrapper, and to use the public AUTO code first on simplified Parker and frozen-source reference problems.

## 12.2 Collocation nodes are not weak-quadrature nodes

AUTO-07p is a close algorithmic precedent, but it does not implement Eq. (9). At a collocation node it enforces the strong ODE defect,

$$\mathbf{r}_q = \frac{d\mathbf{y}_h}{dr}(r_q) - \mathbf{f}(r_q, \mathbf{y}_h, \boldsymbol{\theta}) = 0. \quad (45)$$

In the weak formulation, Gaussian weights instead approximate an integral after integration by parts. Both methods use polynomial representations and Gaussian nodes, but their residuals, conditioning, and conservation properties differ. The prototype must label them separately:

1. *strong collocation*: closest to AUTO-07p and simplest for a regular first-order BVP;

2. *conservative weak residual*: preferred when exact element flux balance and discontinuous source structure dominate; and
3. *existing finite-volume residual*: required when the target is the current EXHALE discrete fixed point.

### 12.3 ESTER: the primary astrophysical reference

ESTER computes steady two-dimensional models of rapidly rotating stars with spectral expansions and nonlinear iterations. Its radial representation uses Chebyshev polynomials, its angular representation uses spherical harmonics, and its coordinate mapping follows the distorted stellar surface. Newton iterations are used when realistic microphysics makes simpler fixed-point iterations inadequate [15].

ESTER demonstrates that a global astrophysical structure calculation can couple smooth field coefficients, equation-of-state and opacity tables, radiative transport, and large-scale flow without evolving the full transient. The transferable lessons are coefficient and equation scaling, continuation from simple stellar models, mapped multidomain coordinates, and separate monitoring of each physical equation. Its direct spectral discretization should not be copied blindly for an EXHALE wind: a sonic transition, outward characteristics, and localized ionization fronts give the wind a different mathematical structure.

### 12.4 Dedalus and KADATH as independent design checks

Dedalus is an open spectral PDE framework developed for astrophysical, geophysical, and fluid-dynamical applications [1]. Its nonlinear boundary-value interface rewrites symbolic equations and constraints as a root problem and applies Newton iterations. It is useful for rapidly testing a simplified continuum formulation of advection, diffusion, column equations, and boundary conditions. It is not the recommended production dependency, because reimplementing EXHALE chemistry, radiation, tables, and boundary logic in a second framework would create a new physical code rather than a reduced front end to EXHALE.

KADATH is a public C++ spectral library for nonlinear systems in theoretical physics and numerical relativity [10]. Its multidomain representations and equation-system abstraction are relevant if EXHALE later requires separate smooth domains around a sonic point or narrow thermal front. Its principal applications are elliptic equilibrium systems, so it supplies software-structure and spectral-convergence precedents rather than a ready advection–diffusion solver.

### 12.5 Recommended inheritance for EXHALE

The codes should be used as references, not combined as dependencies. The recommended inheritance is:

1. retain the EXHALE residual and microphysics ownership from Formulation A;
2. adopt AUTO-07p’s piecewise-polynomial collocation, adaptive defect control, restart metadata, and pseudo-arclength continuation for the continuum phase;
3. adopt ESTER’s scaling, progressive activation of microphysics, and mapped smooth-domain practices;
4. use Dedalus only for small independent equation and boundary-condition experiments; and

5. use KADATH as a reference if multidomain spectral geometry becomes necessary.

This survey therefore changes no physical acceptance criterion. It strengthens the case for a deterministic adaptive collocation and continuation stage, but the final solution must still pass the unmodified EXHALE residual or be labeled explicitly as a continuum weak solution.

### 13. Implementation plan in the EXHALE tree

No production hydrodynamic path should be changed for the first prototype. Add an isolated module and driver so that the experiment cannot alter a normal EXHALE run.

| proposed component                                         | responsibility                                                                                                                                                                                 |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| src/modules/time_step/<br>reduced_steady_state.f90         | spline map, coefficient transforms, exact mass-continuity elimination, residual scaling, and reduced merit                                                                                     |
| src/utis/reduced_steady_driver.f90                         | standalone input, initialization, optimization and continuation control, checkpoints, and diagnostics                                                                                          |
| src/modules/time_step/<br>steady_conservation_residual.f90 | later implementation of element balances, strong collocation defects, and Gaussian weak quadrature; absent from Phase A                                                                        |
| src/modules/functions/<br>species_transport_flux.f90       | common evaluation of advective, molecular-diffusive, and eddy-diffusive element fluxes, factored from the existing update so that time integration and residual assembly use identical physics |
| tests/reduced_steady/                                      | analytic winds, saved EXHALE states, basis and quadrature refinement gates                                                                                                                     |

None of the components in this table exists in the repository: they are the proposal of this memo, not a description of the tree. The files the memo cites elsewhere — `steady_residual.f90`, `steady_newton.f90`, `post_process_adv.f90`, `wae_difeq.f90`, `binary_element_diffusion.f90` — are the existing ones.

The existing full residual evaluator currently accepts and mutates a species state. A reduced wrapper should expose a pure logical interface:

$$(\mathbf{a}, \boldsymbol{\theta}) \mapsto (\mathbf{r}, \text{status}, \text{diagnostics}), \quad (46)$$

where identical inputs produce identical outputs. The status must distinguish invalid state, chemistry nonconvergence, boundary violation, table-range error, and a successfully evaluated large residual. Returning one generic penalty for all failures creates flat regions that are difficult for both optimization and MCMC.

Useful diagnostic output for every accepted iteration includes

- total and equation-wise scaled norms;
- unscaled integrated mass, momentum, and energy imbalance;
- element flux variation, separated advective and diffusive contributions, and ion-stage source-sum error when transport is active;
- radius and equation of the largest residual;

- $\dot{M}$ , sonic radius, maximum temperature, and outer Mach number;
- chemistry iteration counts and fallback events;
- minimum density, pressure, temperature, and species fraction; and
- basis size, element count, and quadrature order.

Checkpoint coefficient vectors rather than only sampled profiles. A checkpoint must also store knot positions, coordinate mapping, residual scales, active physics, boundary configuration, and the source revision. Coefficients have no meaning without that metadata.

## 14. Validation program

### 14.1 Gate 1: analytic and semi-analytic problems

Begin with physics that has a known branch:

1. isothermal Parker wind in a point-mass potential;
2. hydrostatic isothermal atmosphere with  $\dot{M} = 0$ , using a formulation that does not divide by velocity; and
3. adiabatic wind with prescribed smooth heating.

Measure profile error, sonic radius, conservation residual, and convergence under basis refinement. The Parker test is especially important because an optimizer can find subsonic breeze and supersonic branches that have small local residuals but do not satisfy the intended critical condition.

### 14.2 Gate 2: projection of an existing EXHALE state

Project a trusted full-grid state onto the spline basis without solving. This answers whether the proposed basis can represent the profile at all. Record profile errors and the change in the full EXHALE residual caused only by projection. If projection alone produces a large residual, the knots or variables must be changed before optimizer behavior is studied.

### 14.3 Gate 3: hydrodynamics with frozen microphysics

Hold heating, cooling, and composition fixed from a reference state and solve only the reduced hydrodynamics. This isolates basis geometry, scaling, boundary conditions, and sonic selection. It is a numerical gate, not a physically self-consistent final result.

### 14.4 Gate 4: fully coupled discrete residual

Enable equilibrium chemistry and radiation in the same path used by the current steady solver. Compare against a full-grid JFNK solution when one exists. Acceptance requires agreement in the physical profiles and a small unmodified EXHALE residual; agreement in  $\dot{M}$  alone is insufficient.

## 14.5 Gate 5: conservative species transport

Activate transport processes one at a time before combining them:

1. with diffusion disabled, verify constant elemental fraction and the advective element flux implied by the hydrodynamic solution;
2. with  $v = 0$  and fixed temperature, recover the appropriate diffusive equilibrium or barometric separation profile;
3. impose the base abundance and verify zero outer diffusive flux without suppressing outward advective escape;
4. require  $r^2 J_X$  to be constant for every conserved element and verify the same result from face fluxes and integrated weak residuals;
5. compare a reduced simultaneous diffusion solution with the converged production outer iteration, while labeling any changed energy closure;
6. in ion-advection tests, require the sum of reaction sources to conserve nuclei and the summed stage residual to match the element residual; and
7. compare conditional outward integration with the guarded `post_process_adv.f90` result only inside the latter's documented validity region.

Repeat the tests in advection-dominated and diffusion-dominated limits. A matching abundance profile without matching total flux is not sufficient.

## 14.6 Gate 6: weak formulation

For Algorithm B, demonstrate independently:

1. quadrature convergence at fixed basis and elements;
2. element refinement at fixed basis;
3. basis refinement at fixed element layout;
4. small held-out strong residual;
5. stable cumulative columns and heating/cooling integrals; and
6. convergence toward Algorithm A as both discretizations are refined, or a documented continuum-versus-finite-volume difference.

## 14.7 Gate 7: robustness and branch search

Use cold, transonic, Wind-AE, and perturbed starts. A unique physical branch should converge to the same state. Distinct solutions must each satisfy the same boundary conditions and residual tolerance before being labeled multiple physical branches. A mode caused by chemistry history, residual weighting, or an unconverged quadrature is numerical.

## 14.8 Gate 8: short dynamic check

Map the reduced solution to the production grid and evolve it briefly with the ordinary EXHALE time step. The state should remain stationary within the measured residual scale. This check does not replace the residual test, and it must not be extended into an hours-long regeneration merely to reproduce an existing product.

## 15. Failure modes and safeguards

| failure                          | symptom                                                         | safeguard                                                               |
|----------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------|
| over-smooth basis                | low broad merit but a missed heating or ionization layer        | held-out residual; adaptive local knots; inspect source profiles        |
| quadrature aliasing              | residual small only at fitted nodes                             | increase $Q$ without refitting; use independent nodes                   |
| bad residual scaling             | one equation improves while another remains physically wrong    | print equation-wise unscaled balances and largest rows                  |
| false Bayesian certainty         | narrow coefficient posterior controlled by an arbitrary $\beta$ | label it a search distribution; vary scale and basis                    |
| chemistry memory                 | repeated identical coefficients give different merit            | pure evaluator; restore state; tighten equilibrium convergence          |
| wrong branch                     | smooth subsonic breeze accepted as a transonic wind             | impose and verify both sonic conditions when physically required        |
| incompatible boundary conditions | nonzero residual floor localized at a boundary                  | analyze characteristics and physical BCs; do not smear the error        |
| global polynomial ringing        | oscillations around a narrow transition                         | local B-splines or spectral elements; knot refinement                   |
| coefficient degeneracy           | large posterior correlations and slow MCMC                      | remove redundant basis directions; orthogonalize or whiten coefficients |
| invalid finite-volume comparison | weak solution differs from HLLC fixed point                     | report the discretization distinction; refine both formulations         |

Transport introduces three additional failure modes. First, separately discretized advective and diffusive divergences can mimic an elemental source; assemble and test one total face flux. Second, all  $S$  ion-stage equations plus an element equation produce a rank-deficient system; retain only  $S - 1$  independent stage constraints. Third, setting the outer *total* species flux to zero blocks physical escape; set only the diffusive contribution to zero when that is the intended boundary condition. If species enthalpy transport or frictional heating is omitted, record the approximation rather than silently claiming conservation for a more complete multicomponent model.

A scalar  $\chi^2$  can conceal a physically wrong compromise. In particular, if boundary and interior constraints are incompatible, an optimizer or MCMC sampler will distribute error according to the chosen weights. It cannot make the equations compatible. Physical correctness is determined from the individual balances, boundary characteristics, and critical-point topology, not from a small scalar merit alone.

## 16. Decision points before implementation

The following choices materially change the solver and should be fixed before production implementation:

1. **Target equation.** Exact current finite-volume fixed point (Formulation A), continuum weak solution (Formulation B), or both in sequence.
2. **Physical branch.** Outward transonic wind, subsonic L1-truncated flow, or hydrostatic atmosphere. One parameterization should not silently force all three.
3. **Chemistry closure.** Local equilibrium eliminated at each trial, or advected ion fractions represented as additional fields.
4. **Transport closure.** Production operator splitting with outer co-convergence, or a simultaneous conservative residual for elemental diffusion and ion-stage advection. Also state whether diffusive enthalpy and frictional heating belong to the target model.
5. **Boundary treatment.** Hard constraints built into the basis versus soft residuals with stated scales.
6. **Role of probability.** Annealed global search, numerical discretization uncertainty, uncertain physical parameters, or observational inference. These interpretations require different covariance models.

For the current EXHALE structure, the recommended first decision is narrow: target the discrete residual, solve an outward wind with equilibrium chemistry, hard-enforce base temperature and steady mass continuity, retain the production boundary routine, and use deterministic trust-region least squares. This prototype directly tests the central hypothesis: that a smooth 20–30 dimensional state manifold can reach a fixed point that the full  $3N$ -variable solver has difficulty approaching. Elemental diffusion should then be added first by reproducing the production outer co-convergence, followed by a simultaneous conservative transport residual. Self-consistent ion-stage advection is a later extension because it changes the chemistry closure and coefficient count. Once this discrete prototype is stable, introduce AUTO-style adaptive collocation and pseudo-arclength continuation without changing the production microphysics implementation.

## 17. Acceptance criteria for a successful prototype

The prototype is successful only if all of the following are demonstrated:

1. repeated coefficient evaluations are deterministic to the requested tolerance;
2. the reduced solution satisfies mass continuity by construction and both momentum and energy residuals under the unmodified production diagnostic;
3. when species transport is active, total element fluxes satisfy their conservative residuals, transport boundary conditions are met, and ion-stage source sums conserve nuclei;
4. results are stable under basis refinement and initialization;
5. the inferred sonic structure and outer characteristic state are physically consistent with the selected domain;

6. the final profile remains stationary in a short production-code evolution;
7. any difference from a reference solution is separated into physics, boundary, basis, quadrature, and discretization contributions; and
8. measured wall-clock cost, residual-call count, and chemistry failures are reported without extrapolating from the hours-long marching runtime.

MCMC is an optional success criterion. It becomes useful after the deterministic solver works, either because several valid branches exist or because uncertain physical inputs and observations define a genuine posterior. Failure of a random-walk chain to mix is not evidence that the reduced steady formulation is wrong; failure of the independently evaluated physical residual is.

## 18. Summary

The proposed method replaces the full radial state by smooth spline expansions of velocity and temperature plus one mass-loss parameter. Density is eliminated exactly through steady mass continuity, while the existing EXHALE equilibrium chemistry determines composition and pressure conditionally. The first solver should minimize the current full-grid finite-volume residual, because it gives an unambiguous comparison with the production fixed point. A later weak spectral-element solver can integrate conservative momentum and energy balances with Gaussian quadrature and may require far fewer radial evaluations.

Advection and diffusion fit the same framework. Bulk hydrodynamic advection, viscosity, and conduction are already part of the current discrete residual. Elemental diffusion requires a small set of additional abundance coefficients and a residual for the total advective–diffusive element flux. Ion-stage advection requires either independent softmax spline fields or a conditional upstream integration. The current operator-split diffusion and post-processed ion advection must not be described as if they were already simultaneous JFNK residual blocks.

The production workflow is more robust to sharp structure, has an unambiguous conservative finite-volume fixed point, and provides a fallback when Newton fails. Its main costs are the long transient, the high-dimensional JFNK state, and split treatment of composition transport. The reduced workflow removes many redundant search directions and supports direct continuation, but adds basis error, explicit sonic-branch responsibility, and a possible distinction from the production discretization. The recommended architecture therefore uses the reduced solve to enter the full-grid JFNK basin and retains ordinary EXHALE as the final residual diagnostic and short stability test.

Among the surveyed public codes, AUTO-07p is the closest numerical reference for the one-dimensional reduced boundary-value problem because it combines piecewise-polynomial collocation, adaptive meshes, boundary and integral constraints, and continuation. ESTER supplies the closest astrophysical precedent for a nonlinear spectral solve coupled to realistic microphysics. Dedalus is useful for independent continuum prototypes, while KADATH becomes relevant only if a multidomain spectral representation is required. These precedents support the deterministic solver architecture but do not replace the EXHALE conservation and fixed-point validation gates.

The scalar residual may be written in a likelihood form, but nonlinear least squares is the appropriate first solver. Annealed MCMC or sequential Monte Carlo can search for multiple branches; posterior MCMC is meaningful when a

physical error model or observational likelihood is present. In every case, equation-wise conservation, boundary consistency, deterministic chemistry, basis refinement, quadrature refinement, and held-out residuals remain the acceptance criteria.

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